

Plant Hunt: Three Plants, Two Systems, One Food Factory

Name: _____ Date: _____

I can find the root system and shoot system of three different plants, and show where and how photosynthesis happens in each one.

Your mission: find and sketch **1 tree**, **1 bush** (shrub), and **1 patch of grass**.

- Stay inside the boundary your teacher sets.
- **Look, don't pick.** Don't pull up, break, or eat any plant. If you're not sure a plant is safe to touch, don't touch it.
- Roots are underground, so **don't dig**. Draw the roots you *predict* are there using **dashed lines**.
- Bring a clipboard and a pencil. A green pencil crayon helps with the photosynthesis step.

Read Photosynthesis in a Nutshell

Photosynthesis is how a plant makes its own food. In the **green parts** of a plant (mostly the leaves), tiny **chloroplasts** use green **chlorophyll** to capture **light energy**. The plant uses that energy to turn **water** (pulled up from the roots through the xylem) and **carbon dioxide** (taken in from the air) into **glucose**, a sugar that feeds the whole plant. **Oxygen** is released into the air as a by-product.

WHAT GOES IN



Light energy
from the sun



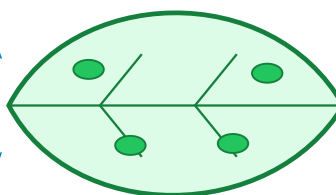
Water
roots → up the xylem



Carbon dioxide
from the air, into the leaf

PHOTOSYNTHESIS

happens in the leaf



green dots = chloroplasts
(chlorophyll captures the light)

WHAT COMES OUT



Glucose
food, sent round
the plant by phloem



Oxygen
released into the air

light energy + water + carbon dioxide → glucose + oxygen

Model What a Finished Sketch Looks Like

Here is a dandelion, sketched and labelled the way you will do your three plants. Use it as your guide. Your drawings don't need to be art. They need to be **clear and labelled**.

WORD BANK: PLANT SYSTEMS AND PARTS

root system

shoot system

roots

stem (a tree's stem = trunk)

branches

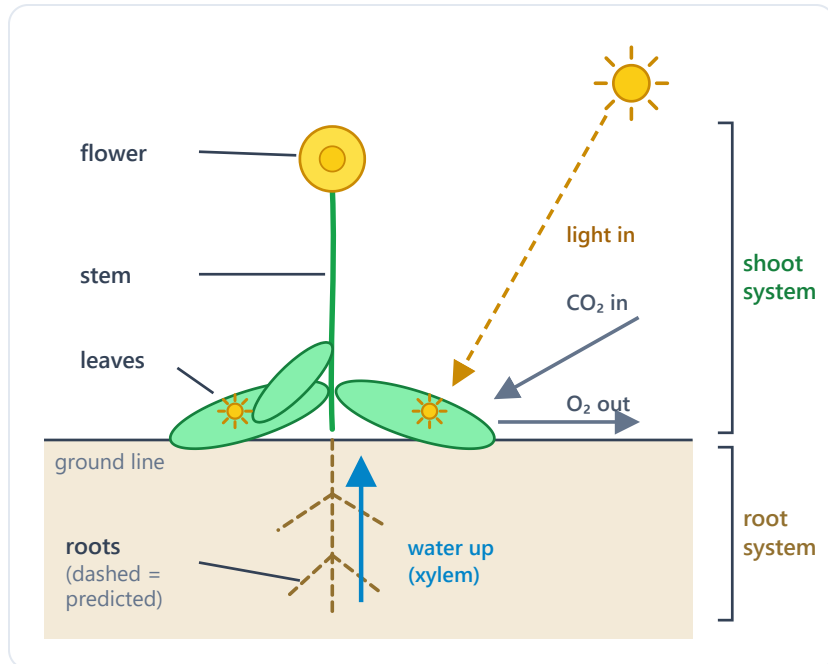
leaves (needles and grass blades are leaves)

flower

seeds / fruit / cones

xylem (water up)

phloem (food around)



Every sketch needs:

1. A **ground line**. It's already printed in each drawing box.
2. The **shoot system** above the ground, drawn from what you see.
3. The **root system** below the ground, drawn as a **dashed prediction**.
4. A bracket and label for **both systems**.
5. At least **4 parts** labelled from the word bank.
6. **Shade green** every part that does photosynthesis and add the sun mark ☀.
7. Photosynthesis **arrows**: light in, CO₂ in, O₂ out, and water up from the roots.

--- light — gases (CO₂ in, O₂ out) — water --- predicted roots ☀ photosynthesis happens here

No coloured pencils? Use a plain pencil and write "light", "CO₂", "O₂", and "water" on your arrows.

Predict Before You Go Outside

Which of your three plants (tree, bush, or grass) do you predict has the **biggest root system** compared with the size of its shoot system? Explain what a root system does that makes you think so.

After the hunt: was your prediction right? _____



Plant 1 • A Tree

Draw the whole tree small enough to fit, with its top near the top of the box.

Kind of tree (if you know): _____ Where I found it: _____

- ☐ ground line ☐ shoot system ☐ root system (dashed) ☐ both systems bracketed ☐ 4+ parts labelled
☐ green + sun mark ☐ photosynthesis arrows

ABOVE GROUND: WHAT YOU SEE

BELOW GROUND: WHAT YOU PREDICT

root system

shoot system

roots

stem (trunk)

branches

leaves / needles

seeds / fruit / cones

xylem

phloem

1. A tree's trunk is its stem. Look closely at the bark. Does the trunk do much photosynthesis? Explain what you saw that helped you decide.

2. This tree's leaves might be 10 m or more above its roots. Explain how water gets from the roots up to the leaves, and why the leaves need it.



Plant 2 • A Bush (Shrub)

Look low, near the ground, at where the stems begin.

Kind of bush (if you know): _____ Where I found it: _____

- ☐ ground line ☐ shoot system ☐ root system (dashed) ☐ both systems bracketed ☐ 4+ parts labelled
☐ green + sun mark ☐ photosynthesis arrows

ABOVE GROUND: WHAT YOU SEE

BELOW GROUND: WHAT YOU PREDICT

root system

shoot system

roots

stems

branches

leaves

flower

seeds / fruit / berries

xylem

phloem

1. How are this bush's stems different from the tree's trunk? Name one way they are the *same* in the job they do.

2. Find the side or part of the bush that gets the most sunlight. Are there more leaves there, or fewer? Use what photosynthesis needs to explain why.



Plant 3 • Grass

Zoom in: draw 2 or 3 grass plants **large**, not the whole lawn.

Kind of grass (if you know): _____ Where I found it: _____

- ☐ ground line ☐ shoot system ☐ root system (dashed) ☐ both systems bracketed ☐ 4+ parts labelled
☐ green + sun mark ☐ photosynthesis arrows

ABOVE GROUND: WHAT YOU SEE

BELOW GROUND: WHAT YOU PREDICT

root system

shoot system

roots

stem

leaves (blades)

seed head

xylem

phloem

1. Grass has no trunk and no branches. Which part of the grass does the photosynthesis? How could you tell by looking?

2. A lawnmower cuts off the top half of every grass blade. Predict what happens to how much glucose the grass can make right after mowing. Explain using the flowchart.

Compare Your Three Plants

Use your sketches to fill in the table.

	What its stem looks like	Where photosynthesis happens	The root system I predicted (deep? wide? a mat?)
Tree			
Bush			
Grass			

3. All three plants look very different. What is the **same** about where and how they make their food? Use at least two words from the flowchart.

THINK IT THROUGH

A big tree can weigh more than a car. A classmate says: "*The tree got all that mass by taking food out of the soil.*"

4. Do you agree or disagree? Use the photosynthesis flowchart to explain where a tree's food (glucose) really comes from, and what the soil *does* give it.

One part of photosynthesis or plant systems I'm still not sure about: _____